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Physics-Constrained Rectified Flow for Climate Downscaling

A post-sampling projection makes rain non-negative and matches the coarse mean, and matched three-year training shows no measurable gain from the penalties.

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1. Overview

Objective: Downscale coarse ERA5 precipitation to a fine grid with a generative model. Its rain should stay non-negative, and its domain mean should match the coarse input.

Limit of current practice: Sharp generative downscalers can produce negative rain, and so can bicubic interpolation. Our CorrDiff reimplementation has the best PSNR on ERA5 2020 (28.38) with one seed per arm, yet 45% of the cells of its ensemble-mean precipitation are negative. In the three-year run its rate is 0.108, near bicubic at 0.119.

KEY TAKEAWAY
Put guarantees in the inference path, and use the training loss only to shape the field.

2. Method

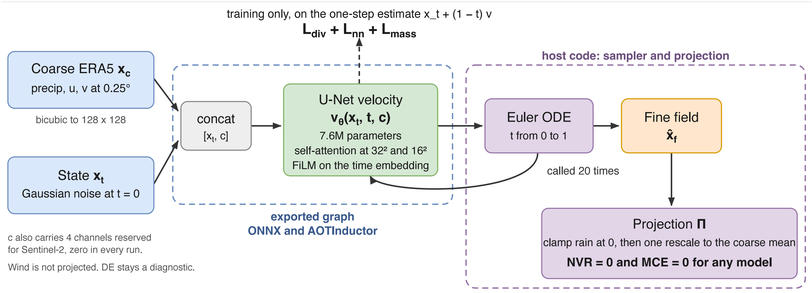


Figure 1. PC-RF as trained and served. The 7.6M-parameter U-Net is the only exported graph. The Euler sampler and the projection run in host code.

Training: Rectified flow learns a velocity from noise to data, and three penalties act on a one-step clean-field estimate.

$$\mathcal{L}_{\mathrm{RF}} = \mathbb{E}\left\|v_\theta(x_t,t,c) - (x_1 - x_0)\right\|_2^2, \qquad \hat{x}_1 = x_t + (1-t)\,v_\theta(x_t,t,c) \tag{1}$$

$$\mathcal{L} = \mathcal{L}_{\mathrm{RF}} + \lambda_1\mathcal{L}_{\mathrm{div}} + \lambda_2\mathcal{L}_{\mathrm{nn}} + \lambda_3\mathcal{L}_{\mathrm{mass}}, \qquad \lambda_1 = \lambda_2 = \lambda_3 = 1 \tag{2}$$

Inference: Each of eight ensemble members runs twenty Euler steps. The projection then clamps rain at zero and applies one rescale per sample, so the domain mean matches the coarse mean $\bar{p}_c$ over land and sea. This domain-level rescaling follows Hess et al. (2022), who rescale the precipitation of a trained GAN once after training.

$$\Pi(p) = \max(p,0) \cdot \frac{\bar{p}_c}{\mathrm{mean}\left(\max(p,0)\right)} \tag{3}$$

The wind is not projected. A divergence-free projection needs a Poisson solve over the whole domain, and near-surface convergence is physically real.

3. Data and Setup

- The **coarse input** is ERA5 precipitation and 10 m u, v at 0.25° over 40–50°N, 10°W–5°E, at 00, 06, 12 and 18 UTC.
- The **fine target** is ERA5-Land precipitation at 0.1° (about 8 to 11 km), resampled to 128×128, with ocean cells set to zero. No fine wind target exists, so the wind target is the upsampled coarse wind.
- The **splits** are chronological 80/10/10 on 2020 (1171 training samples) and on 2019–2021. The 2020 test set runs from late November to 31 December.
- The **baselines** are bicubic and our reimplementations of SRCNN, SRGAN, DDPM and CorrDiff, scored without the projection.
- The **main run** trained 300 epochs at batch 32 with AdamW (learning rate 2×10^{-4}) on one A100, about 6.2 GPU-hours.
- Each arm used **one training seed**, and the Sentinel-2 channels were zero in every run.

REFERENCE CAVEAT
ERA5-Land precipitation is interpolated from ERA5 forcing over a different accumulation window. The perceptual scores therefore measure agreement with an interpolant and cannot establish downscaling skill. The validity metrics never use the reference.

4. Results

4.1 Synthetic mechanism check

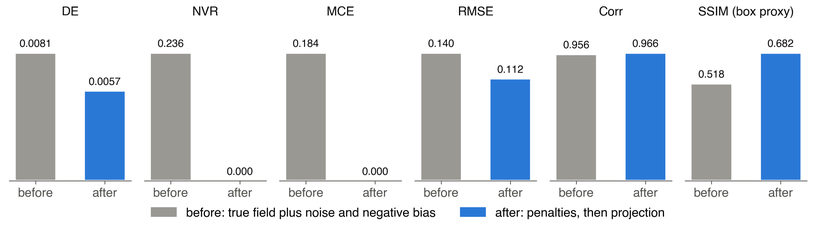


Figure 2. The check uses one constructed 128×128 field with exact physics, and no network is trained. The "before" field is the truth plus correlated noise and a negative bias. The projection drives NVR and MCE to zero, and the penalties cut divergence error by about 30%. The fidelity gain cannot be split between the two steps.

4.2 ERA5 2020: only projected models are exactly valid

Model	SSIM↑	PSNR↑	CRPS↓	DE	NVR↓	MCE↓
Bicubic	0.885	26.19	1.356	0.0648	0.101	0.0003
SRCNN	0.833	27.74	1.461	0.0738	0.053	11.14
SRGAN	0.855	27.03	1.506	0.0700	0.177	13.22
DDPM	0.656	26.18	1.609	0.0715	0.826	7.50
CorrDiff	0.864	**28.38**	**1.092**	0.0679	0.451	6.48
RF-base	0.912	26.39	1.188	0.0643	**0**	**0**
PC-RF	**0.914**	26.59	1.149	0.0644	**0**	**0**

Table 1. Shaded rows carry the projection. RF-base trained 200 epochs at batch 16, and the other learned models trained 300. CRPS uses the 8 members, and the other metrics use their mean. SSIM, PSNR and CRPS also average two wind channels whose target is the upsampled coarse wind. MCE is a relative error of the domain mean, not a water amount, and DE, a diagnostic, is in standardized grid units.

4.3 One year at a matched budget

Arm (200 epochs, batch 16)	SSIM↑	PSNR↑	CRPS↓	DE
Full PC-RF	0.912	26.54	1.161	0.0634
No divergence penalty	0.912	**26.58**	**1.146**	0.0649
No non-negativity penalty	0.913	26.55	1.160	0.0644
No mass penalty	**0.915**	26.50	1.166	0.0639
RF-base (no penalties)	0.913	26.40	1.186	0.0643

Table 2. The penalty arms tie RF-base on SSIM and lead it slightly on PSNR and CRPS. With one seed per arm, that lead cannot be separated from run-to-run variation. The same RF-base checkpoint gives the Table 1 row, scored there with a fresh 8-member sample.

4.4 Three years at a matched budget: a tie

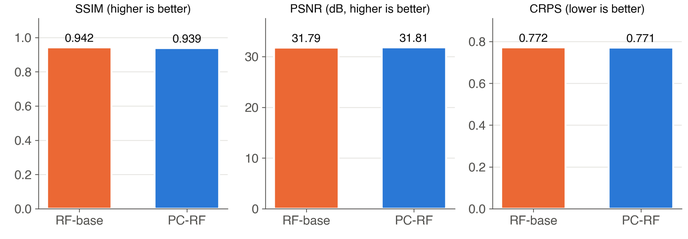


Figure 3. Every arm trained 100 epochs on 2019–2021. RF-base matches PC-RF on every perceptual score, and both keep NVR and MCE at zero. These scores mix rain and wind.

RESULT
The penalties add no measurable quality at a matched three-year budget. The validity guarantee belongs to the projection.

5. What the Checks Certify

- **Reference-free**: NVR and MCE need only the generated field and its coarse input.
- **Certificates**: after the projection both are zero for any model, up to float32 rounding, so they report the inference step and cannot rank two projected models.
- **Diagnostic**: On ERA5, a latitude sign error makes DE measure a deformation term, not divergence, and it carries no guarantee. The reference wind is upsampled coarse wind, whose own DE is 0.0648 on 2020 and 0.0567 on 2019–2021.
- **Necessary, not sufficient**: a field can pass every check and still be wrong.

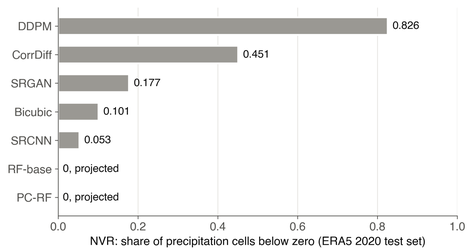


Figure 4. Share of precipitation cells below zero on the ERA5 2020 test set. Every model scored without the projection fails the gate, bicubic included. Bicubic's negatives come from interpolation overshoot at rain edges, and a clamp would remove them.

RECOMMENDATION
Run NVR and MCE first as a cheap, reference-free gate, and keep DE beside it as a diagnostic.

6. From PyTorch to Production

Export the network, not the sampler: The exported graph is one velocity call, $v_\theta(x_t, t, c)$. The Euler loop and the projection stay in host code, mirrored in Python and Rust.

Serving on one A100: One request (20 steps plus the projection, batch 1) takes 91.7 ms in eager PyTorch, 43.3 ms with AOTInductor, 49.1 ms with `torch.compile`, and 65.5 ms with ONNX Runtime, which copies through the host every step. The weights are random in the trained architecture.

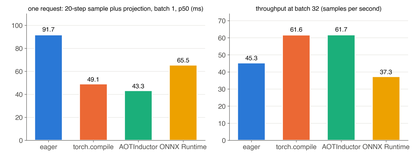


Figure 5. One request and throughput at batch 32 on an A100-SXM4-40GB. With TF32 off, `torch.compile` and AOTInductor match eager within 2×10^{-6}, and NVR stays at zero on every backend. The Rust service and Ray Serve were tested on a demo-scale export, and every model here trained on one GPU.

7. Limitations and Next Steps

- The reference cannot show downscaling skill. An independent target with its own precipitation would test the mass penalty.
- The projection matches the coarse mean over land and sea, a relation the reference breaks. The rescale needs at least one positive cell.
- We used one domain, one test season, and one seed per arm, with no extremes evaluation.
- Baselines were scored without the projection, which any of them could apply.
- On ERA5, a sign error makes the divergence penalty and DE measure a deformation term, not divergence. The synthetic check, in grid coordinates, is unaffected.

8. Resources

Repo github.com/arunshar/pcrf-aias2026-talk
Slides github.com/arunshar/pcrf-aias2026-talk
Video youtu.be/xyiUB-ehX8g
Paper github.com/arunshar/pcrf-aias2026-talk

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